



Afraz Khan

AI ENGINEER

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SUMMARY

AI Engineer based in Islamabad, specializing in LLMs, multi-agent systems, and computer vision. I design and deploy scalable ML solutions — from multi-agent LLM pipelines and semantic search to OCR-powered document automation — for healthcare and FinTech. My work has slashed operational costs by €40,000, boosted query accuracy by 40%, and cut customer support response times by 75%.

EXPERIENCE

AI Engineer

Present

OphyCare · Islamabad, Pakistan (Hybrid · Full-time)

- Built an end-to-end medical scribing pipeline converting patient-doctor interactions into structured clinical data.
- Automated population of 10+ mandatory medical forms with context-aware Generative AI, eliminating hours of manual entry per provider.
- Deployed multi-agent systems where specialized agents collaborate to resolve complex healthcare workflows including eligibility verification.
- Integrated advanced GenAI solutions to streamline the most time-consuming parts of clinical documentation.

Python

LangGraph

OpenAI

Deepgram

pgvector

FastAPI

AI / ML Engineer

Apr 2025 — May 2025

EDGE C Pvt. Ltd · Lahore, Pakistan (Hybrid · Full-time)

- Drove a 40% uplift in query accuracy over legacy approaches using Neo4j knowledge graphs and MongoDB vector embeddings.
- Built LangChain orchestration pipelines delivering a 35% improvement in answer relevance for complex financial queries.
- Developed gRPC microservices with Redis caching achieving sub-200 ms end-to-end retrieval latency.
- Collaborated with data engineering and product teams to define financial ontology schemas and optimize embedding strategies.

Python

LangChain

Neo4j

MongoDB

Redis

gRPC

AI Engineer

Sep 2024 — Jan 2025

Ace Money Transfer · Kharian, Pakistan (Full-time)

- Designed and implemented ML models for critical financial operations and transaction analytics.
- Engineered robust data preprocessing pipelines for complex, high-volume transactional datasets.

Python

Scikit-learn

PyTorch

Data Pipelines

AI Intern

Prior

Rev9 Solutions · Jhelum, Pakistan (Internship)

- Engineered ML models to address client business challenges across industries under senior AI engineers.
- Transformed raw data into structured formats for model training and validation.
- Integrated solutions into production environments alongside senior AI engineers.

Python

Scikit-learn

TensorFlow

EDUCATION

BS, Computer Science

CGPA 3.8 / 4.0

University of Punjab · Lahore, Pakistan

SKILLS

LLMs & Agents	OpenAI, LangGraph, LangChain, Multi-Agent Systems, RAG, Semantic Search, Prompt Engineering
Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn, Deep Learning, NLP, BERT / Transformers, Hugging Face
Computer Vision	PaddleOCR, Tesseract, YOLOv8, OpenCV, DeepFace, Image Processing, Document Analysis
Backend & APIs	Python, FastAPI, Flask, gRPC, REST APIs, Docker, Microservices
Data & Infrastructure	Neo4j, pgvector, MongoDB, Redis, PostgreSQL, Vector DBs, Knowledge Graphs
MLOps & Tools	Git, Deepgram, Roboflow, Labellmg, Model Optimization, Data Visualization

SELECTED PROJECTS

Eligibility Verification Agent

Mar 2026 — Present

Automated insurance eligibility for healthcare

- AI agent automating the full eligibility pipeline — clearinghouse inquiry → EMR updates → physician sign-off.
- Eliminates manual back-and-forth between clinical staff, clearinghouses, and EMR systems.

Sadiq Scribe

Jul 2025 — Feb 2026

AI medical scribing & form automation

- Transcribes patient-doctor audio and auto-populates 10+ mandatory medical forms.
- End-to-end pipeline: real-time STT → clinical entity extraction → intelligent form-field mapping.

SMR — Smart Medical Record

Oct 2025 — Feb 2026

Clinical Q&A with citation support

- Hybrid retrieval combining semantic search with structured lookups surfaces citation-backed answers.
- Physicians query current and historical appointment data with proper source attribution.

Intellectus

Apr 2025 — May 2025

Hybrid knowledge retrieval for FinTech

- Dual-module system combining graph-based and semantic search — 40% uplift in query accuracy.
- Neo4j for entity relationships, MongoDB vectors for unstructured content, LangChain orchestration (+35% answer relevance).

KYC — Know Your Customer

Sep 2024 — Jan 2025

End-to-end verification system for FinTech

- Slashed annual operational costs by €40,000 with a full-stack KYC solution.
- Integrated PaddleOCR and DeepFace for document verification and facial recognition.

Doc Detector & Classifier

Oct 2024 — Nov 2024

Lightweight document analysis for FinTech

- 98% field detection accuracy on identity documents; trained on 5,000+ custom-labeled documents.
- Lightweight YOLOv8n model reduced compute requirements by 70% for edge deployment.

Knowledge Base Q&A

Jan 2024 — Feb 2024

Intelligent information retrieval engine

- NLP-powered Q&A system that cut customer support response times by 75%.
- Hybrid TF-IDF + BERT retrieval; Flask API handling 1,000+ concurrent queries at sub-100ms.

